

```
QEMU (Win-1)
C:\WINDOWS\system32\cmd.exe
Reply from 10.0.0.20: bytes=32 time=2ms TTL=128
Reply from 10.0.0.20: bytes=32 time=3ms TTL=128
Reply from 10.0.0.20: bytes=32 time=2ms TTL=128
Reply from 10.0.0.20: bytes=32 time=3ms TTL=128

Ping statistics for 10.0.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 10.0.0.20

Pinging 10.0.0.20 with 32 bytes of data:

Reply from 10.0.0.20: bytes=32 time=3ms TTL=128
Reply from 10.0.0.20: bytes=32 time=1ms TTL=128
Reply from 10.0.0.20: bytes=32 time=3ms TTL=128
Reply from 10.0.0.20: bytes=32 time=2ms TTL=128

Ping statistics for 10.0.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>
```

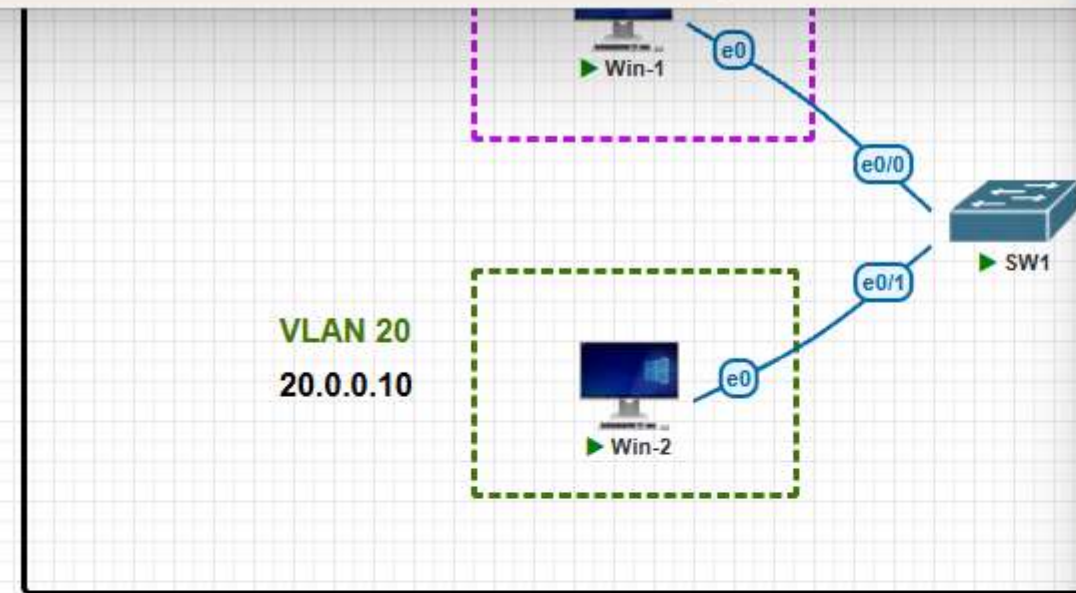
QEMU (Win-2)

Capturing from Local Area Connection [Wireshark 1.10.14 (v1.10.14-0-g825f971 from master-1.10)]

File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help

Filter: Expression... Clear Apply Save

No.	Time	Source	Destination	Protocol	Length	Info
30	32.3006290	aa:bb:cc:00:01:10	CDP/VTP/DTP/PAGP/UDDP	90	Dynamic Trunking Protocol	
31	34.0007190	aa:bb:cc:00:01:10	Spanning-tree-(for-STP	60	Conf. Root = 32768/20/aa:bb:cc:00:01:00 Cost = 0 Port	
32	36.0006040	aa:bb:cc:00:01:10	Spanning-tree-(for-STP	60	Conf. Root = 32768/20/aa:bb:cc:00:01:00 Cost = 0 Port	
33	38.0007570	aa:bb:cc:00:01:10	Spanning-tree-(for-STP	60	Conf. Root = 32768/20/aa:bb:cc:00:01:00 Cost = 0 Port	
34	40.0006410	aa:bb:cc:00:01:10	Spanning-tree-(for-STP	60	Conf. Root = 32768/20/aa:bb:cc:00:01:00 Cost = 0 Port	
35	42.0007020	aa:bb:cc:00:01:10	Spanning-tree-(for-STP	60	Conf. Root = 32768/20/aa:bb:cc:00:01:00 Cost = 0 Port	
36	44.0002180	aa:bb:cc:00:01:10	Spanning-tree-(for-STP	60	Conf. Root = 32768/20/aa:bb:cc:00:01:00 Cost = 0 Port	
37	46.0004580	aa:bb:cc:00:01:10	Spanning-tree-(for-STP	60	Conf. Root = 32768/20/aa:bb:cc:00:01:00 Cost = 0 Port	
38	48.0004670	aa:bb:cc:00:01:10	Spanning-tree-(for-STP	60	Conf. Root = 32768/20/aa:bb:cc:00:01:00 Cost = 0 Port	
39	50.0007370	aa:bb:cc:00:01:10	Spanning-tree-(for-STP	60	Conf. Root = 32768/20/aa:bb:cc:00:01:00 Cost = 0 Port	
40	50.9496380	20.0.0.20	20.0.0.10	ICMP	74	Echo (ping) request id=0x0200, seq=3328/13, ttl=128 (re
41	50.9497340	20.0.0.10	20.0.0.20	ICMP	74	Echo (ping) reply id=0x0200, seq=3328/13, ttl=128 (re
42	51.9532640	20.0.0.20	20.0.0.10	ICMP	74	Echo (ping) request id=0x0200, seq=3584/14, ttl=128 (re
43	51.9533780	20.0.0.10	20.0.0.20	ICMP	74	Echo (ping) reply id=0x0200, seq=3584/14, ttl=128 (re
44	52.0008510	aa:bb:cc:00:01:10				



Frame 34: 60 bytes on wire (480 b)

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

0000	01	80	c2	00	00	00	aa	bb	cc	00
0010	03	00	00	00	00	00	80	14	aa	bb
0020	00	00	80	14	aa	bb	cc	00	01	00
0030	02	00	0f	00	00	00	00	00	00	00

```
QEMU (Win-4)
C:\WINDOWS\system32\cmd.exe
Reply from 20.0.0.10: bytes=32 time=2ms TTL=128
Reply from 20.0.0.10: bytes=32 time=2ms TTL=128
Reply from 20.0.0.10: bytes=32 time=3ms TTL=128
Reply from 20.0.0.10: bytes=32 time=3ms TTL=128

Ping statistics for 20.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

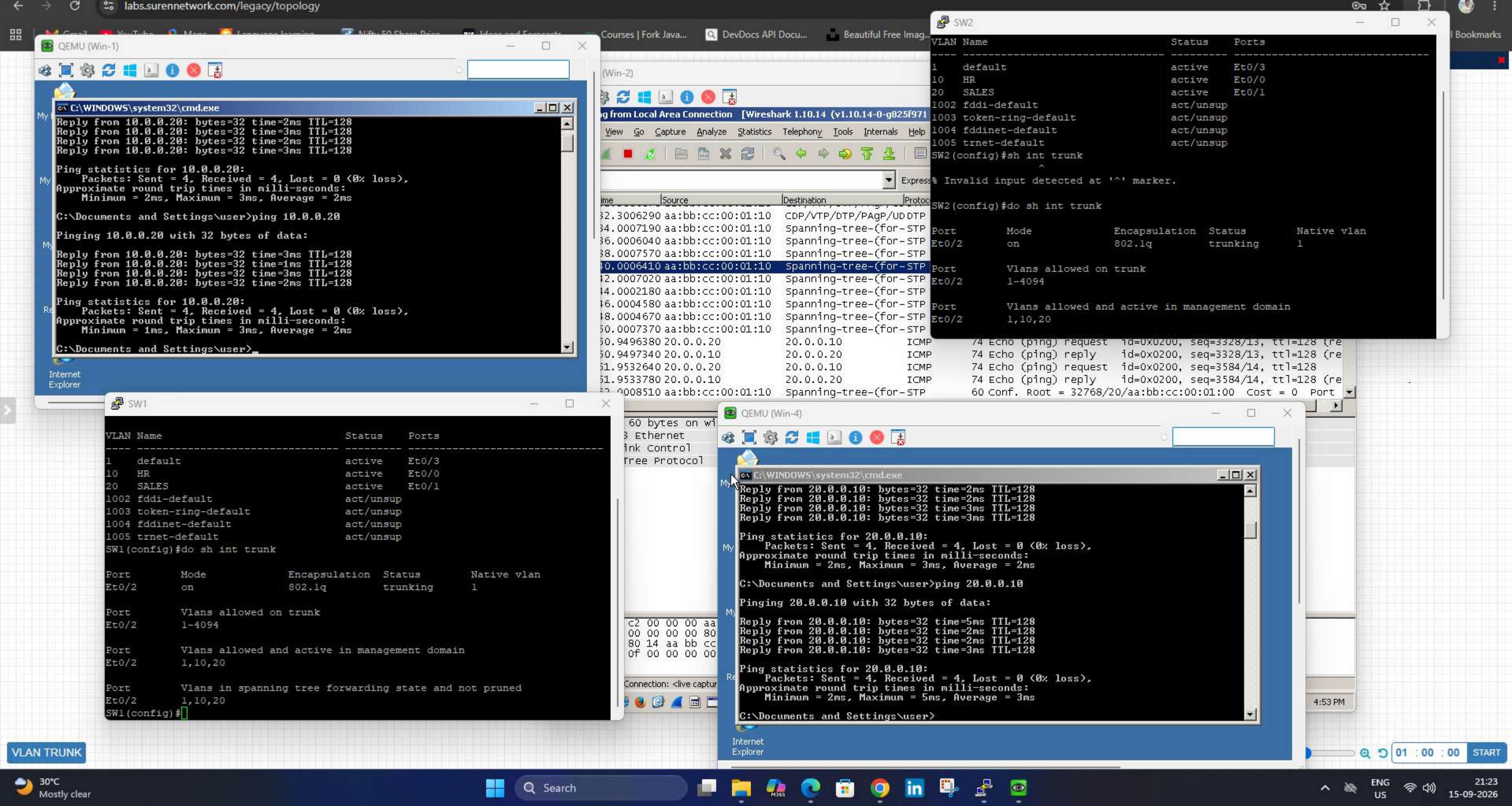
C:\Documents and Settings\user>ping 20.0.0.10

Pinging 20.0.0.10 with 32 bytes of data:

Reply from 20.0.0.10: bytes=32 time=5ms TTL=128
Reply from 20.0.0.10: bytes=32 time=2ms TTL=128
Reply from 20.0.0.10: bytes=32 time=2ms TTL=128
Reply from 20.0.0.10: bytes=32 time=3ms TTL=128

Ping statistics for 20.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 5ms, Average = 3ms

C:\Documents and Settings\user>
```



```
C:\WINDOWS\system32\cmd.exe
Reply from 10.0.0.20: bytes=32 time=2ms TTL=128
Reply from 10.0.0.20: bytes=32 time=3ms TTL=128
Reply from 10.0.0.20: bytes=32 time=2ms TTL=128
Reply from 10.0.0.20: bytes=32 time=3ms TTL=128

Ping statistics for 10.0.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 10.0.0.20

Pinging 10.0.0.20 with 32 bytes of data:
Reply from 10.0.0.20: bytes=32 time=3ms TTL=128
Reply from 10.0.0.20: bytes=32 time=1ms TTL=128
Reply from 10.0.0.20: bytes=32 time=3ms TTL=128
Reply from 10.0.0.20: bytes=32 time=2ms TTL=128

Ping statistics for 10.0.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>
```

Wireshark 1.10.14 (v1.10.14-0-g825f971)

Time	Source	Destination	Protocol
32.3006290	aa:bb:cc:00:01:10	CDP/VTP/DTP/PagP/UD	DTP
34.0007190	aa:bb:cc:00:01:10	Spanning-tree-(for-	STP
36.0006040	aa:bb:cc:00:01:10	Spanning-tree-(for-	STP
38.0007570	aa:bb:cc:00:01:10	Spanning-tree-(for-	STP
40.0006410	aa:bb:cc:00:01:10	Spanning-tree-(for-	STP
42.0007020	aa:bb:cc:00:01:10	Spanning-tree-(for-	STP
44.0002180	aa:bb:cc:00:01:10	Spanning-tree-(for-	STP
46.0004580	aa:bb:cc:00:01:10	Spanning-tree-(for-	STP
48.0004670	aa:bb:cc:00:01:10	Spanning-tree-(for-	STP
50.0007370	aa:bb:cc:00:01:10	Spanning-tree-(for-	STP
50.9496380	20.0.0.20	20.0.0.10	ICMP
50.9497340	20.0.0.10	20.0.0.20	ICMP
51.9532640	20.0.0.20	20.0.0.10	ICMP
51.9533780	20.0.0.10	20.0.0.20	ICMP
52.0008510	aa:bb:cc:00:01:10	Spanning-tree-(for-	STP

SW2

VLAN	Name	Status	Ports
1	default	active	Et0/3
10	HR	active	Et0/0
20	SALES	active	Et0/1
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
SW2(config)#sh int trunk
% Invalid input detected at '^' marker.

SW2(config)#do sh int trunk

Port      Mode      Encapsulation  Status      Native vlan
Et0/2     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Et0/2     1-4094

Port      Vlans allowed and active in management domain
Et0/2     1,10,20

60 Conf. Root = 32768/20/aa:bb:cc:00:01:00 Cost = 0 Port
```

SW1

VLAN	Name	Status	Ports
1	default	active	Et0/3
10	HR	active	Et0/0
20	SALES	active	Et0/1
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
SW1(config)#do sh int trunk

Port      Mode      Encapsulation  Status      Native vlan
Et0/2     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Et0/2     1-4094

Port      Vlans allowed and active in management domain
Et0/2     1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Et0/2     1,10,20

SW1(config)#
```

```
C:\WINDOWS\system32\cmd.exe
Reply from 20.0.0.10: bytes=32 time=2ms TTL=128
Reply from 20.0.0.10: bytes=32 time=2ms TTL=128
Reply from 20.0.0.10: bytes=32 time=3ms TTL=128
Reply from 20.0.0.10: bytes=32 time=3ms TTL=128

Ping statistics for 20.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 20.0.0.10

Pinging 20.0.0.10 with 32 bytes of data:
Reply from 20.0.0.10: bytes=32 time=5ms TTL=128
Reply from 20.0.0.10: bytes=32 time=2ms TTL=128
Reply from 20.0.0.10: bytes=32 time=2ms TTL=128
Reply from 20.0.0.10: bytes=32 time=3ms TTL=128

Ping statistics for 20.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 5ms, Average = 3ms

C:\Documents and Settings\user>
```

VLAN TRUNK

01 : 00 : 00 START